

Research Methodology in CS (CSEG3060_3)

Lecture 4: Scope and Objectives of Research: Defining Scope and Objectives

Unit 1: Introduction to Research Problem • Mapped Course Outcome: CO2

Lecture 4: Scope and Objectives of Research — Defining Scope and Objectives

Course: Research Methodology in Computer Science (CSEG3060_3) **Program:** B.Tech (Computer Science)
Semester: 5 **Unit:** 1 — Introduction to Research Problem **Mapped Course Outcome:** CO2 **Estimated Reading Time:** 45–55 minutes

1. Introduction and Lecture Roadmap

In the preceding lectures of Unit 1, we examined the *meaning* of a research problem, its *significance*, the *sources* from which research problems originate, and the *criteria and characteristics* that distinguish a good research problem from a poorly formulated one. We also discussed the most common errors researchers commit when selecting a topic.

Once a research problem has been identified and validated, the next critical intellectual task is to translate that problem into a clearly bounded **scope** and a set of well-articulated **objectives**. Without these, even a brilliant problem statement cannot be operationalized into a tractable study. Lecture 4 focuses exclusively on this translation step.

In Computer Science research, where problems are often highly technical, interdisciplinary, and resource-intensive (for example, training large language models or conducting large-scale vulnerability assessments), the discipline of defining scope and objectives becomes especially consequential. A precisely bounded study yields reproducible results, whereas an unbounded one drifts into ambiguity.

By the end of this lecture, the learner should be able to:

- Distinguish clearly between the **scope** of a study and its **objectives**.
 - Apply established frameworks (such as SMART) to formulate research objectives.
 - Recognize delimitations and limitations in a CS research context.
 - Critically evaluate a given research proposal for the quality of its scope and objectives.
 - Convert a high-level research problem into bounded, achievable objectives relevant to computing research.
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2. Key Concepts at a Glance

The following terms form the conceptual vocabulary of this lecture.

Concept	Working Definition
Research Scope	The boundaries of a study — what will and will not be covered.
Research Aim (Goal)	The broad, long-term intention of the research.
Research Objective	A specific, measurable, achievable outcome that contributes to the aim.
Research Question (RQ)	A precise interrogative form of an objective.
Hypothesis	A testable proposition that operationalizes an objective in deductive research.
Delimitation	An intentional boundary imposed by the researcher.
Limitation	A constraint imposed externally (by data, time, resources).
Operationalization	The process of converting a conceptual objective into measurable indicators.

These are not synonyms. Conflating them is a frequent origin of methodological weakness.

3. Detailed Conceptual Exposition

3.1 Defining the Scope of Research

3.1.1 What "Scope" Means

The **scope** of research is the perimeter inside which the entire inquiry will be conducted. It answers the question: "*What exactly am I studying, where, when, on whom (or on what), and under what conditions?*"

In formal project documents (synopses, proposals, theses), the scope is typically expressed through the following dimensions:

1. **Domain or Subject Boundary** — the subdomain of Computer Science (for example, *natural language processing, network security, human-computer interaction, distributed systems*).
2. **Theoretical Boundary** — the set of theories, models, or frameworks consulted (for example, *information retrieval theory, game theory, complexity theory*).
3. **Population or Sampling Boundary** — the datasets, users, software artifacts, hardware platforms, or system events under examination.
4. **Geographical Boundary** — the geographic or institutional context (for example, *an IoT deployment in a single smart-campus testbed*).
5. **Temporal Boundary** — the time horizon of the study (for example, *traffic logs collected between January and June 2024*).
6. **Methodological Boundary** — the methods permitted (for example, *only empirical benchmarks; no theoretical proofs*).

3.1.2 The "Funnel" Visualization of Scope

A useful mental model is the **funnel of specificity**:

Research Domain → Research Area → Research Topic → Research Problem → Research Scope → Objectives
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Each level narrows the inquiry. The scope is the next-to-last level; objectives are the smallest, most concrete formulations.

3.1.3 Delimitations versus Limitations

A **delimitation** is a boundary *deliberately* chosen by the researcher — often because exploring everything is impossible. Examples:

- "The study considers only supervised classifiers; unsupervised and semi-supervised methods are excluded."
- "The proposed algorithm is benchmarked on English-language corpora only, not multilingual inputs."

A **limitation**, by contrast, is a boundary *imposed externally*. Examples:

- "The available GPU compute (single A100) restricts model size."
- "Access to only one medical institution restricts patient diversity in the dataset."

Distinguishing the two signals rigor. Delimitations are choices; limitations are constraints. Both must be acknowledged, but only delimitations are arguable.

3.1.4 Common Errors When Defining Scope

- **Scope Creep** — gradual, uncontrolled expansion of the study after initiation. In software engineering research, this often manifests as adding new modules, datasets, or comparison systems midway.
 - **Over-Scoping** — attempting to address too many sub-problems at once.
 - **Under-Scoping** — carving out a niche so narrow that the result has no generalizable or publishable contribution.
 - **Implicit Scope** — leaving boundaries unstated, leaving the reader to infer them.
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3.2 Defining the Objectives of Research

3.2.1 The Hierarchy: Aim → Objectives → Tasks

Aim and objectives should not be stated as the same sentence at the same level of abstraction.

Level	Example (CS Context)
Aim	To improve the robustness of deep learning models against adversarial perturbations in image classification.
Objective 1	To design a novel adversarial training algorithm that reduces attack success rate by at least 15% on benchmark datasets.
Objective 2	To evaluate the trade-off between clean accuracy and adversarial robustness using standardized metrics such as robust accuracy.
Objective 3	To validate the proposed method across at least three representative attacks (FGSM, PGD, C&W).
Objective 4	To compare the proposed method against three state-of-the-art baselines under identical experimental conditions.

Notice how the aim is essentially non-testable, while each objective is independently testable.

3.2.2 The SMART Framework for Objectives

Objectives in research are widely formalized through the **SMART** mnemonic (a guideline popularized by George Doran in 1981 and now standard in methodology training):

Objective = Specific $\cap$ Measurable $\cap$ Achievable $\cap$ Relevant $\cap$ Time-bound

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Translated into CS research practice:

- **Specific** — the objective identifies *what* will be done, to *what*, in *which* context.
- **Measurable** — there exists a metric or criterion by which success can be judged. For CS, this usually means accuracy, F1-score, AUC, latency (in ms), throughput (in requests/second), energy consumption (in joules), or a qualitative rubric.
- **Achievable** — feasible given available datasets, compute, time, and expertise.
- **Relevant** — aligned with the stated aim and contributes to the field.
- **Time-bound** — schedulable within the research window (often 6, 12, or 24 months).

3.2.3 Types of Objectives

Objectives may be classified along two useful axes.

(a) By generality:

- **General (Primary) Objectives** — the umbrella intent. Usually one statement.
- **Specific (Secondary) Objectives** — the operational breakdowns that collectively satisfy the general objective.

(b) By cognitive intent (drawing on Bloom's revised taxonomy):

The cognitive level of an objective must match its actual deliverable.

Cognitive Level	Verb Examples	Typical CS Deliverable
Remember	identify, list, recall	taxonomy of attacks
Understand	describe, summarize	literature synthesis
Apply	implement, deploy	prototype system
Analyze	compare, examine	benchmark study
Evaluate	assess, justify	comparative empirical study
Create	design, develop, propose	novel algorithm or architecture

A frequent methodological fault is to *state* an objective at the "Create" level but to *conduct* only an "Analyze" level study.

3.2.4 Objectives, Research Questions, and Hypotheses

In qualitative and mixed-methods research, each objective is often rephrased as a **research question (RQ)**.

Objective: To evaluate the latency impact of edge offloading in mobile cloud computing. **Research Question:** What is the effect of edge offloading on request-response latency under varying network conditions?

In quantitative, positivist research, each RQ is often rephrased as a **hypothesis**, where a hypothesis is a falsifiable statement:

$$H_1: \mu_{\text{offload}} < \mu_{\text{local-execution}} \quad (\text{where } \mu \text{ is the mean response latency})$$

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The conceptual chain is therefore:

Aim $\rightarrow$ Objective $\rightarrow$ Research Question $\rightarrow$ Hypothesis

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Not every CS study uses hypotheses; design science and exploratory research often bypass the hypothesis form and proceed directly from objectives to artifacts.

3.3 The Relationship Between Scope and Objectives

Scope and objectives are not redundant; they are mutually constraining.

- A **scope** that is too wide makes the objectives unfalsifiable.
- **Objectives** that are over-ambitious force the scope to expand beyond feasibility.

A well-formulated research problem requires that the scope be **just wide enough** to make the objectives meaningful, and the objectives be **just narrow enough** to fit the scope.

A useful self-diagnostic is the **Objectives-in-Scope Test**:

For each objective, ask: *"Is there a documented, available, and ethical way to gather the data required to satisfy this objective within the stated scope?"*

If the answer is *no* for any objective, the scope or the objective must be revised.

4. Practical Examples and Case Studies

The following examples and cases demonstrate the principles in CS research contexts.

4.1 Example A — Reformulating a Vague Problem

Initial (poorly formed) problem:

"AI is very useful. We want to use AI for healthcare."

Defined scope:

"This study focuses on deep learning models for chest X-ray classification, restricted to publicly available datasets (CheXpert and NIH ChestX-ray14), evaluated using AUROC and clinical sensitivity, on a single institutional GPU cluster, between the academic year 2024–2025."

Defined objectives:

1. To curate and standardize two benchmark chest X-ray datasets for binary pneumonia detection.
2. To design a transfer-learning pipeline based on a pre-trained convolutional backbone.
3. To evaluate the model using five-fold cross-validation against two state-of-the-art baselines.
4. To interpret model decisions using Grad-CAM visualizations.

Each objective is SMART, internally consistent, and bounded by the stated scope.

4.2 Example B — Research Problem in Network Security

Research problem: Insider threats remain a leading cause of data breaches in enterprises.

Scope:

- User behavioral logs from a single enterprise testbed.
- Six-month observation window.
- Focus on three observable channels: file access, email, login patterns.
- Detection restricted to offline batch processing (not real-time streaming).

Objectives:

1. To engineer a multi-view feature representation combining file-system, email, and authentication logs.
2. To train and tune an ensemble anomaly detector based on isolation forests and recurrent neural networks.
3. To evaluate detection using precision, recall, and F1-score against a labeled ground-truth set.
4. To estimate the operational cost (in terms of false alarms per analyst-shift) of the proposed system.

Hypothesis form (for Objective 4):

$$H_1: \text{False alarms per shift} < 3$$

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This illustrates how scope, objectives, and hypotheses integrate into one coherent research design.

4.3 Case Study — Android Malware Detection

Consider the well-known area of Android malware detection research. A quick literature scan reveals the following pattern in published studies:

- Studies that **clearly delimit** themselves to a specific feature family (e.g., *API-call sequences* or *permissions*) and a specific detection paradigm (e.g., *static analysis only*) consistently produce reproducible results.
- Studies that attempt to "cover all features and all detection methods" tend to underperform on any single benchmark, suffering from over-scoping.

Lesson: **Defining scope narrowly is often the path to stronger empirical evidence.** This is a counterintuitive but reliable methodological lesson.

4.4 Example C — Well-Defined versus Poorly-Defined Objectives (Comparative)

Aspect	Poorly-Defined Objective	Well-Defined Objective
Specificity	"To study deep learning."	"To fine-tune BERT variants for sentiment classification on three product-review datasets."
Measurability	"To improve performance."	"To achieve a macro-F1 score improvement of at least 3 percentage points over the baseline."
Achievability	"To solve general AI."	"To reduce inference latency by 20% via INT8 quantization on the same hardware."
Relevance	"To use big data."	"To mitigate class imbalance in real-time network intrusion detection."
Time-Bound	Implicit.	"Within an 8-month implementation window."

The contrast makes SMART violations concrete and memorable.

5. Common Errors When Defining Objectives

The following are recurrent errors observed in undergraduate and early postgraduate research in CS:

- 1. Conflating activities with outcomes.** "To perform literature survey" is an *activity*; "To identify the three most influential research gaps in federated learning" is an *outcome*. Objectives must be outcomes.
- 2. Multiple objectives in a single sentence.** Compound objectives obscure which deliverable serves which evaluation. Split them.
- 3. Objectives that exceed the scope.** An objective requiring access to proprietary medical records when the scope restricts the study to public datasets is unachievable.
- 4. Objectives that are not testable.** "To deeply understand interpretability" cannot be measured and is therefore not a research objective in the strict sense.
- 5. Objectives that contradict the stated aim.** An aim of "improving efficiency" paired with an objective of "improving accuracy" without acknowledging the trade-off signals weak integration.
- 6. Neglecting the time dimension.** Without a schedule anchor, objectives become aspirations rather than commitments.

6. Summary

This lecture has translated the abstract notion of a research problem into the operational constructs of *scope* and *objectives*. The principal takeaways are:

- **Scope** establishes the *boundaries* of the study — domain, theory, population, geography, time, and methodology. Delimitations are chosen; limitations are imposed.
- **Objectives** are the *measurable outcomes* through which the research aim will be realized. They should follow SMART criteria and align with a clear cognitive level.
- A sound research design requires **mutual consistency** between scope and objectives; each objective must be satisfiable within the declared boundaries.
- In CS research, where artifacts (algorithms, systems, datasets) often constitute the contributions, objectives typically lead to one or more artifacts, each accompanied by an evaluation plan.
- The intellectual discipline of writing tight scope statements and SMART objectives is, in itself, a strong indicator of research quality and is among the first things examiners and reviewers evaluate.

Defining scope and objectives is therefore not a bureaucratic preliminary — it is the act that converts intellectual curiosity into executable research.

7. Exam Preparation Tips and Practice Questions

7.1 Preparation Tips

- **Memorize the dimensions of scope.** Examiners frequently ask students to enumerate them from memory.
- **Be fluent with SMART.** Convert any poorly formed objective into SMART form during revision; this builds speed.
- **Practice the Aim → Objective → RQ → Hypothesis chain.** Be ready to construct this chain for any given problem in 10 minutes.
- **Distinguish delimitations from limitations** — a recurring one-mark and two-mark question.
- **Know Bloom's taxonomy verbs.** Pair each CS artifact (model, benchmark, framework) with its appropriate cognitive level.

7.2 Practice Questions

Short Answer Questions (2–3 marks each):

1. Differentiate between *delimitation* and *limitation* in the context of CS research. Provide one example of each from a study on cloud security.
2. List the five elements of the SMART framework and rewrite the following objective in SMART form: "*To study machine learning algorithms.*"
3. Why must the cognitive level of an objective match the actual deliverable of the study? Illustrate with a CS example.

Descriptive Questions (5–10 marks each):

4. Take the following research problem and, in approximately 200 words, define the scope and write three SMART objectives: "*Online learning platforms face challenges in student dropout prediction.*"

5. Discuss with examples the common errors researchers make while defining research scope. How does *scope creep* manifest in software-engineering research projects?
6. Convert the following aim into a hierarchical structure (one aim, four specific objectives, corresponding research questions, and two hypotheses where applicable): "*To improve the privacy of federated learning systems.*"

Application / Higher-Order Question (10–15 marks):

7. **Critique exercise.** A student submits the following objective in a research proposal: "*To study various deep learning models comprehensively for image classification to see which is best.*" Identify at least five methodological weaknesses in this objective. Rewrite it into three well-formed SMART objectives consistent with a defined scope in image classification research.

End of Lecture 4 Notes.